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Who decides? Positioning stakeholders in a decision-making process via network modeling and analysis: Application to regional trains decarbonization

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ABSTRACT

Understanding and controlling stakeholder influence in public–private decision-making remains a significant challenge for public authorities. This study proposes a methodology for diagnosing, modeling, and reorganizing stakeholder contributions within complex decision-making processes. The methodology integrates five steps: stakeholder identification, stakeholder network modeling, network analysis, stakeholder clustering, and a reorganization phase that aligns stakeholder involvement with the specific objectives of each decision stage. The methodology is applied to public–private network data to reveal influence patterns, dominant stakeholders, and structural coalitions. Stakeholders are identified, and their interactions are formalized using a weighted Design Structure Matrix (DSM). Network indicators are computed to characterize interaction intensity and connectivity. Clustering algorithms are applied to identify clusters of closely interacting stakeholders, which serve as the basis for reorganizing stakeholder involvement across decision stages. We applied the methodology to the decarbonization of French regional diesel trains. Data were collected through interviews conducted with multiple regional transport authorities. The results show an intense concentration of decisional weight among industrial stakeholders and higher-level public institutions, indicating governance vulnerabilities stemming from asymmetric access to information and influence. A validation phase was conducted with a regional decision-maker (Corse), confirming the practical relevance of the proposed stakeholder reorganization and supporting its applicability to real-world decision processes. The study demonstrates that dynamically structuring stakeholder involvement can mitigate risks, enhance decision robustness, and assist public authorities in managing private-sector influence. While illustrated through a specific case study, the methodology is generalizable to any multi-actor public–private decision-making context.

1. Introduction

Understanding how industrial stakeholders influence decision-making processes is a longstanding concern in public decision-making studies. As Potters & van Winden noted, "Providing policymakers and legislators with information is often asserted to be one of the most important means by which interest groups influence the policymaking process." (Potters and van Winden, 1992). However, while this observation highlights the centrality of information flows and informal influence mechanisms, existing modeling approaches rarely successfully capture these dynamics comprehensively. In this paper, we develop a methodology for diagnosing and modeling stakeholder networks, conducting influence analysis, and dynamically reorganizing stakeholder contributions.

This methodology is applied to the case study of the decarbonization of French regional trains, a decision context in which regional authorities act as Authorities for Organizing Transport (AOTs). They manage their train fleets and evaluate technological alternatives based on installation costs, social acceptability, and environmental performance. However, this evaluation process depends substantially on information provided by industrial stakeholders.

Aligned with the Paris Agreement, the transport sector aims for a 55% reduction in greenhouse gas emissions by 2030 compared to 1990 levels (European Climate Law, 2021). In France, 35% of the rail network

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